

PAPER_554: Buoyancy-Stratified Factorial Geometry — Riemann Curvature of the Aether Metric

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CP4 Class: BSFG_{RiemannCurvatureAetherMetricCalculator} (#149)

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Context note: The Aether metric $A_{\mu\nu}$ was introduced in PAPER_392 (CP4 #43) with coupling $\eta = 10^{-22}$ and scalar amplitude T_{s00} . PAPER_416 (CP4 #66) provided the five-component spatial decomposition of $T_{s00}(r)$. This paper combines those results to derive for the first time the full Riemann curvature tensor of the BSFG geometry — the first paper in the BSFG Complete Geometric System series (PAPER_554-558).

Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Riemann Curvature of the Aether Metric, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

The Buoyancy-Stratified Factorial Geometry (BSFG) is defined on a 26-dimensional pseudo-Riemannian manifold $\mathcal{M}^{26}$ equipped with the Aether-perturbed metric $A_{\mu\nu}(r) = g_{\mu\nu} + \varepsilon(r) \delta_{\mu\nu}$, where $\varepsilon(r) = \eta T_{s00}(r) \cos(\pi t_n)$ and $T_{s00}(r)$ is the five-component stellar stress-energy density from PAPER_416. This paper derives the Christoffel symbols, Riemann curvature tensor, Ricci tensor, and Ricci scalar for the 4D slice of BSFG at a field point r . The key result is:

$$R^r_{0r0} \approx \frac{\varepsilon''}{2} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

with $C_{\text{num}} = (M_s c^2 + L_s / c^2) / (4\pi/3)$. At the solar surface $r = R_\odot$: $R^r_{0r0} \approx 1.56 \times 10^{-19} \text{ m}^{-2}$, which is $\approx 4 \times 10^{-26}$ times the Schwarzschild curvature — confirming the Aether contribution is a genuine but ultra-weak geometric perturbation.

§2 The Aether Metric — Previous Results

From PAPER_392 (CP4 #43), the BSFG metric is:

$$A_{\mu\nu} = g_{\mu\nu} + \eta T_{s00} \cos(\pi t_n) \delta_{\mu\nu}$$

In components (4D Minkowski background with signature $+---$):

$$A_{\mu\nu} = \text{diag}(1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon)$$

where $\varepsilon = \eta T_{s00}(r) \cos(\pi t_n)$. From PAPER_416 (CP4 #66), the five-component Ts00 with dominant radial term:

$$T_{s00}(r) = \underbrace{\frac{M_s c^2}{\frac{4}{3}\pi r^3}}_{T_1 \propto r^{-3}} + \underbrace{\frac{L_s}{c^2 \cdot \frac{4}{3}\pi r^3}}_{T_2 \propto r^{-3}} + \underbrace{\frac{\rho_{SCm} v_{SCm}^2}{c^2} + \frac{\rho_A v_{UA}^2}{c^2} + \rho_{sw} v_{sw}^2}_{\text{constant terms}}$$

The radially-dependent numerator is $C_{\text{num}} = (M_s c^2 + L_s/c^2)/(4\pi/3) \approx 4.27 \times 10^{46} \text{ J} \cdot \text{m}$.

§3 Christoffel Symbols

Step 1. Compute $\varepsilon(r)$ and its radial derivatives:

$$\varepsilon'(r) = \frac{d\varepsilon}{dr} = -\frac{3\eta \cos(\pi t_n) C_{\text{num}}}{r^4}, \quad \varepsilon''(r) = +\frac{12\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

Step 2. For a diagonal metric $A_{\mu\mu}(r)$ depending only on $r = x^1$, the non-zero Christoffel symbols of the Levi-Civita connection are:

$$\Gamma_{\mu\mu}^r = -\frac{\partial_r A_{\mu\mu}}{2 A_{rr}} = -\frac{\varepsilon'}{2(-1 + \varepsilon)} \quad (\text{no sum on } \mu)$$

$$\Gamma_{\alpha r}^\alpha = \Gamma_{r\alpha}^\alpha = \frac{\partial_r A_{\alpha\alpha}}{2 A_{\alpha\alpha}} \quad (\text{no sum on } \alpha)$$

Explicitly at leading order in ε :

Symbol	Exact	Leading order
Γ_{00}^r	$-\varepsilon'/(2(-1 + \varepsilon))$	$+\varepsilon'/2$
Γ_{rr}^r	$+\varepsilon'/(2(-1 + \varepsilon))$	$-\varepsilon'/2$
Γ_{0r}^0	$+\varepsilon'/(2(1 + \varepsilon))$	$+\varepsilon'/2$
Γ_{ir}^i	$+\varepsilon'/(2(-1 + \varepsilon))$	$-\varepsilon'/2$
Γ_{ii}^r	$-\varepsilon'/(2(-1 + \varepsilon))$	$+\varepsilon'/2$

§4 Riemann Curvature Tensor

Step 3. Apply the Riemann tensor formula:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$$

The dominant component (tidal force in the radial-temporal plane):

$$R^r{}_{0r0} = \partial_r \Gamma^r_{00} + \Gamma^r_{rr} \Gamma^r_{00} - \Gamma^r_{00} \Gamma^0_{0r}$$

$$= \frac{\varepsilon''}{2} - \frac{(\varepsilon')^2}{2} + O(\varepsilon^3) \approx \frac{\varepsilon''}{2}$$

Step 4. Substituting $\varepsilon'' = 12\eta \cos(\pi t_n) C_{\text{num}}/r^5$:

$$R^r{}_{0r0} = \frac{6\eta \cos(\pi t_n) C_{\text{num}}}{r^5}$$

The higher-order $(\varepsilon')^2$ correction is of order $\eta^2 T_{s00}^2 \sim 10^{-30}$, negligible.

§5 Ricci Tensor and Ricci Scalar

Ricci tensor — applying $R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$ with $SO(3)$ spherical symmetry ($x^2 = y$, $x^3 = z$ contribute equally to $x^1 = r$):

$$R_{00} = R^r{}_{0r0} + R^\theta{}_{0\theta0} + R^\phi{}_{0\phi0} = 3 R^r{}_{0r0}$$

$$R_{rr} = -R^r{}_{0r0} + 2 \left(\frac{\varepsilon''}{2} - \frac{(\varepsilon')^2}{4} \right)$$

Ricci scalar:

$$R = A^{\mu\nu} R_{\mu\nu} = \frac{R_{00}}{A_{00}} + \frac{R_{rr}}{A_{rr}} + \frac{2R_{\theta\theta}}{A_{\theta\theta}}$$

Kretschner scalar (leading order):

$$K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} \approx 12 (R^r{}_{0r0})^2$$

§6 Numerical Values at the Solar Surface

At $r = R_{\odot} = 6.96 \times 10^8 \text{ m}$, $t_n = 0$ (maximum coupling), $\eta = 10^{-22}$:

Quantity	Value	Notes
$\varepsilon(R_{\odot})$	1.27×10^{-2}	Not small — linearization is qualitative at surface
$\varepsilon'(R_{\odot})$	$-5.47 \times 10^{-11} \text{ m}^{-1}$	Aether gradient
$\varepsilon''(R_{\odot})$	$+3.13 \times 10^{-19} \text{ m}^{-2}$	Curvature driver
R^r_{0r0}	$+1.56 \times 10^{-19} \text{ m}^{-2}$	BSFG tidal curvature
R_{scalar}	$\approx +3.0 \times 10^{-19} \text{ m}^{-2}$	de Sitter (gravity-dominant)
K	$\approx 2.9 \times 10^{-37} \text{ m}^{-4}$	Kretschner scalar

For comparison, the Schwarzschild curvature at $R_{\odot}$: $R^r_{0r0}|_{\text{GR}} \approx GM_{\odot}/R_{\odot}^3 \approx 3.95 \times 10^{-7} \text{ m}^{-2}$, giving:

$$\frac{R^r_{0r0}|_{\text{BSFG}}}{R^r_{0r0}|_{\text{GR}}} \approx \frac{1.56 \times 10^{-19}}{3.95 \times 10^{-7}} \approx 3.9 \times 10^{-13}$$

The BSFG Aether curvature is $\sim 10^{12}$ times weaker than GR at the solar surface, consistent with the perturbative assumption $\eta \sim 10^{-22}$ being a tiny coupling.

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow g_{tt} = -(1-2GM/rc^2) \equiv \text{GR}$ in $\varepsilon_{\text{BSFG}} \rightarrow 0$ limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	$\square$ BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow \Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} \times (1 + \varepsilon_{\text{correction}})$	Cassini: $\Delta t / \Delta t_{\text{GR}} = 1 \pm 2.3\text{e-}5$	Cassini/GR 2003	$\square$ Within Shapiro bound
Gravitational wave speed v_{GW}	BSFG: $v_{\text{GW}} = c \times (1 + k_{\eta}^2) \approx c + 10^{-226} \text{ m/s}$	GW150914 / GW170817:	$v_{\text{GW}}/c - 1$	$< 10^{-15}$

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\varphi = \kappa \times \varphi_{\text{GR}} \sim 10^{-6}$ arcsec/century	GR prediction: 43.03"/century; observed: 43.1"	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10^{-6}$ arcsec/ century to Mercury’s perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

§7 Connection to UQFF Framework

The Riemann tensor R^r_{0r0} of the BSFG geometry provides the **geometric encoding of tidal UQFF forces**. The buoyancy force F_U^{bi} represents the differential curvature between interior ($r < R_*$) and exterior ($r > R_*$) regions — a curvature discontinuity at the stellar boundary. The Aether field $\varepsilon(r) \propto r^{-3}$ creates a genuine non-flat geometry whose curvature $\propto r^{-5}$ decays rapidly away from the source, consistent with the UQFF fifth-force measurements reported in PAPER_413-418.

Hub: PAPER_554 (#149) is the first paper in the BSFG series. See PAPER_558 (#153) for the complete geometric system definition and PAPER_555 (#150) for metric compatibility and geodesic equation.